

SAFETY

To prevent serious bodily injury, you should observe the following basic safety precautions when installing, operating or servicing the milling machine.

1. Follow all instructions in the manual.
2. Wear approved industrial safety glasses and safety shoes.
3. Do not wear gloves, long sleeves, long hair, rings, watches, jewelry or other items that could become caught in moving parts.
4. Keep all parts of your body away from moving parts (belts, cutters, gears, etc.)
5. Use proper point of operation safeguarding.

These and other safety precautions are discussed in the American National Standard Institute standard entitled safety requirements for the construction, care, and use of drilling, milling, and boring machines (ANSI B11-8-1983).

This publication is available from:

American National Standards Institute
25 West 43rd Street, 4th floor
New York, NY 10036

Safeguarding for protection at the point of operation can only be designed and constructed when the parameters of the particular operation have been determined. As a result, ANSI B11.8-1983, Section 5.1, states that *"it shall be the responsibility of the employer to provide, and ensure use of, a guard, guarding device, awareness barrier, awareness device, or shield..."*

To assist machine users in designing point of operation safeguarding for their specific machine applications, the Occupational Safety And Health Administration has published a booklet entitled Concepts and Techniques of Machine Safeguarding (O.S.H.A. Publication No. 3067).

This publication is available from:

The Publication Office – O.S.H.A.
U.S. Department of Labor
200 Constitution Avenue, NW
Washington, D.C. 20210

The general purpose point of operation shield provided with this machine and shown in certain illustrations throughout this manual may not be appropriate and cannot be utilized for all possible applications of the machine. Use additional or alternate safeguarding where this shield is not appropriate or cannot be utilized. Note that for purposes of display, the shield has been removed in certain other illustrations in this manual.

- WARNING -

Warning notices are used in this publication to emphasize that hazardous mechanical conditions, voltages, currents, or temperatures exist in this equipment which could cause serious personal injury and/or damage to the equipment.

- CAUTION -

Caution notices are used where equipment might be damaged if care is not taken.

In situations where inattention could cause either personal injury or damage to the equipment, a warning notice is used.

- NOTE -

Notes merely call attention to information that is especially significant in understanding and operating the equipment.

This document is intended for the use of those who install, operate and maintain the milling machine. Although reasonable care has been exercised in the preparation of this manual to make it complete and accurate, this manual does not purport to cover all conceivable problems or applications pertaining to this machine.

SAFETY RECOMMENDATIONS

DO NOT OPERATE EQUIPMENT until you have read and understood the appropriate operator and safety maintenance manuals.

DO NOT OPERATE EQUIPMENT until you have read and understood all machine and control key signs.

DO NOT OPERATE EQUIPMENT for the first time without a qualified instructor. Consult your supervisor when in doubt as to the correct way to perform an operation.

DO NOT OPERATE EQUIPMENT unless proper maintenance has been regularly performed and the equipment is known to be in good working order.

DO NOT ALLOW the operation or repair of equipment by untrained personnel.

WARNING or INSTRUCTION TAGS are mounted on the equipment for your safety and information. Do not remove them.

DO NOT OPERATE EQUIPMENT if any unusual or excessive heat, noise, smoke, or vibration occurs. Report any excessive or unusual vibration, sounds, smoke, or heat as well as any damaged parts.

WEAR SAFETY GLASSES with side shields and SAFETY SHOES with steel toes and oil-resistant soles at all times. When necessary, wear respirator, helmet, and ear muffs or plugs.

DO NOT OPERATE ANY MACHINE while wearing rings, watches, jewelry, loose clothing, neckties, or long hair not contained by a net or shop cap.

DO NOT WEAR GLOVES while operating equipment. Gloves are easily caught in moving parts.

REMOVE ANY LOOSE PARTS OR TOOLS left on machine or in the work area before operating the machine. Always check the machine and work area for loose tools and parts, especially after work has been completed by maintenance personnel.

REMOVE CHUCK WRENCHES before starting the machine.

NEVER OPERATE A MACHINE after taking strong medication, using non-prescription drugs or consuming alcoholic beverages.

SAFEGUARD THE CUTTING ZONE (“point of operation”). Use standard, general purpose safeguards when possible. Use special safeguards when required.

PROTECT YOUR HANDS. Stop the spindle completely before changing tools.

PROTECT YOUR HANDS. Stop the spindle completely before loading or unloading a workpiece.

DO NOT REMOVE CHIPS with hands. Use a hook or similar device and make certain that all machine movements have ceased.

DO NOT ADJUST tooling, workpieces or coolant hoses while the machine is running.

PROTECT YOUR HANDS. Stop the spindle completely before taking measurements.

PROTECT YOUR HANDS. Stop the spindle completely before opening safeguards or covers.

NEVER REACH around a safeguard.

PROTECT YOUR HANDS. Stop the machine before changing or adjusting belts, pulleys or gears.

PROTECT YOUR HANDS. Keep hands and arms clear of spindle start switch when changing tools.

PROTECT YOUR EYES AND THE MACHINE. Never use a compressed air hose to remove chips.

KEEP WORK AREA WELL LIGHTED. ask for additional light if needed.

DON'T SLIP. Keep your work area clean and dry. Remove chips, oil and obstacles.

NEVER LEAN ON your machine. Stand away when the machine is running.

MAKE CERTAIN that you are clear of any “pinch points” created by moving slides before starting the machine.

PREVENT OBJECTS from flying loose. Securely clamp and locate workpiece. Use stop blocks where necessary. Keep clamps clear of cutter path.

PREVENT CUTTER BREAKAGE. Use correct table feed and spindle speed for the job. Reduce feed and speed if you notice unusual noise or vibration.

PREVENT CUTTER BREAKAGE. Rotate spindle in clockwise direction for right-hand tools, counterclockwise for left-hand tools. Use the correct tool for the job.

PREVENT WORKPIECE and cutter damage. Never start the machine when the cutter is in contact with the workpiece.

DO NOT USE worn or defective tools. Use the proper size and type of tool for the task at hand.

KEEP ROTATING CRANKS AND HANDWHEELS well lubricated and maintained. Do not remove safety springs.

CERTAIN MATERIALS, such as magnesium, are highly flammable in dust and chip form. See your supervisor before working with these materials.

PREVENT FIRE. Keep flammable liquids and materials away from work area and hot chips.

PREVENT MACHINE from moving unexpectedly. Disengage power feed when not being used (manual machines only).

PREVENT MACHINE from moving unexpectedly. Always start machine in manual mode.

UNLESS OTHERWISE NOTED, all operating and maintenance procedures are to be performed by one person. To avoid injury to yourself and others, be sure that all personnel are clear of the machine when opening or closing the coolant guard door and any access covers.

INSTALLATION AND USE OF SAFEGUARDS

Both American National Standard B11.8 and OSHA Section 1910.212 assign responsibility for point of operation safeguarding of milling machines to the employer/user. Therefore, to prevent serious injury resulting from the rotating cutter, flying chips, or splashing coolant, point of operation safeguarding should be used on milling machines to the greatest extent practicable.

This booklet provides basic information for the installation and use of the general purpose safeguard. It also contains the names of several manufacturers of other types of point of operation safeguarding for vertical milling machines.

Remember, point of operation safeguarding is your responsibility as the employer/user. You are in the best position to evaluate your safeguarding needs and ensure that the proper safeguards are installed and used.

- CAUTION -

A safety shield is supplied for protection from chips and coolant with every machine.

The chip and coolant shields have been designed and are custom manufactured with the highest clear impact material commercially available: polycarbonate (G.E. Lexan). It has an impact strength 5 to 10 times greater than acrylic (plexiglass) or butyrate (UVEX) materials, thereby offering the greatest protection for our customers.

Some of the new "easy to dispose of" coolants and/or cutting oils contain chemicals harmful to polycarbonate. These chemicals are: Mono-ethanolamine, Di-ethanolamine, Tri-ethanolamine and the combination thereof. These chemicals may significantly reduce the impact strength of the shield within days, and could destroy the entire shield in weeks.

Use of use of coolants and/or cutting oils containing these chemicals will void the warranty on your safety shield, and could cause injury to your workers.

- WARNING -

This safeguard DOES NOT take the place of any other safety practice or safety equipment.

YOU MUST ALWAYS wear safety glasses and safety shoes.

YOU MUST ALWAYS stop the spindle of the machine completely before changing or adjusting the workpiece, fixture, or tool.

YOU MUST NEVER wear gloves, long sleeves, long hair, rings, watches, neckties, jewelry or other loose items.

GENERAL PURPOSE SAFEGUARDS

There is no single safeguard which can match the versatility of the Series I machine. As a result, you will find that the general purpose safeguard, Figure I.I, like all safeguards, will be suitable for some operations, but not for others. Carefully analyze the operation to be performed before deciding whether this safeguard is suitable. Adjust the safeguard to suit your special requirements. If you find that it is not suitable for a particular application, you should use an alternate form of protection.

Installation for Machines with R-8 Spindle Taper

If your Series I standard milling machine was manufactured on or after January 1, 1983, the manufacturer has drilled and tapped two holes in the nose cap of the spindle to be used for mounting the guard (the two untapped holes serve to remove the nose cap with a spanner wrench).

1. Place mounting ring "B" underneath top of guard "D".
2. Place two socket head cap screws "A" through the holes in ring, and hand-start them into the threaded holes in the nose cap until hand tight.
3. Align guard to be square with table of machine (unless angular mounting is desired).
4. Tighten screws with a 3/16" Allen wrench.

Retrofits

If your Series I standard machine was manufactured prior to January 1, 1983, you must retrofit as follows:

1. Manually tap the existing spanner wrench holes with a .250"-20 bottoming tap.

- CAUTION -

There is no need to drill new holes, nor deepen the existing holes. Attempts at either practice may result in spindle damage.

2. Proceed to mount the guard as indicated above.

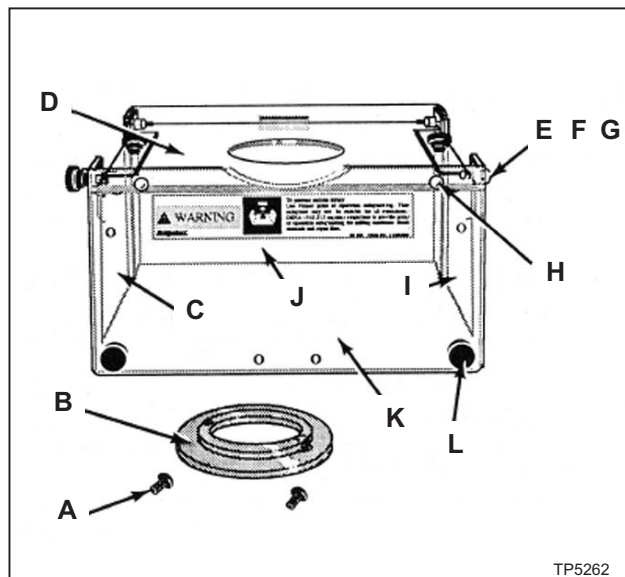


Figure I.I - General Purpose Safeguard Assembly